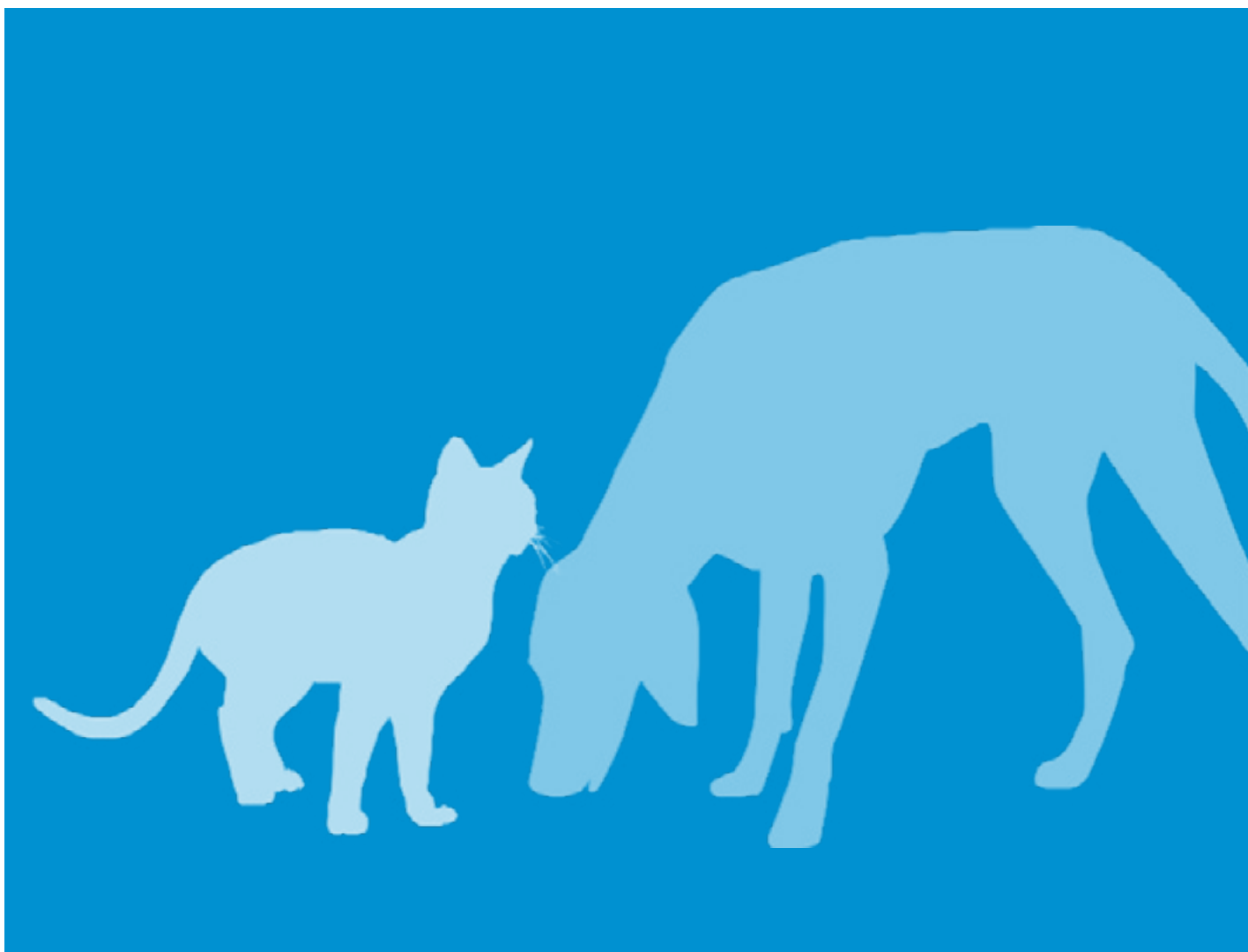


Diagnosing, Staging, and Treating Chronic Kidney Disease in Dogs and Cats

Chronic kidney disease (CKD) is diagnosed based on evaluation of all available clinical and diagnostic information in a stable patient. Following diagnosis of CKD, the IRIS Board recommends using serum creatinine or SDMA (ideally both) to stage CKD with substaging based on assessment of arterial blood pressure and proteinuria.



Diagnosing, Staging, and Treating Chronic Kidney Disease in Dogs and Cats

Step 1: Diagnose CKD

Clinical signs and physical examination findings worsen with increasing severity of kidney disease

Clinical presentation

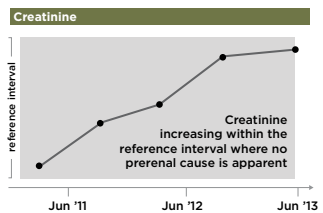
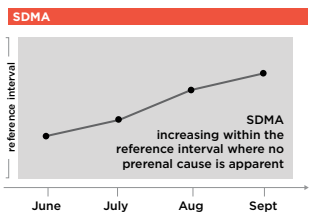
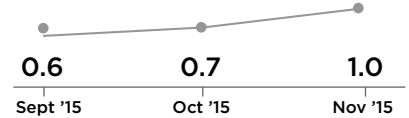
Consider age, sex, breed predispositions, and relevant historical information, including medication history, toxin/toxicant exposure, and diet. Can be subclinical in early stage CKD. Signs may include polyuria, polydipsia, weight loss, decreased appetite, lethargy, dehydration, vomiting, and bad breath.

Physical examination findings

Can be normal in early stage CKD. Findings may include palpable kidney abnormalities, evidence of weight loss, dehydration, pale mucous membranes, uremic ulcers, evidence of hypertension, i.e., retinal hemorrhages/detachment.

To diagnose Stage 1 and early Stage 2 CKD

One or more of these diagnostic findings:

- 1  
- 2 Persistent increased SDMA* >14 µg/dL
- 3 Abnormal kidney imaging  
- 4 Persistent renal proteinuria
UPC >0.5 in dogs; UPC >0.4 in cats 

See www.iris-kidney.com for more detailed staging, therapeutic, and management guidelines.

OR

To diagnose more advanced CKD (late Stage 2-4)

Both of these diagnostic findings:

Increased creatinine and SDMA concentrations

Creatinine

SDMA

Results of both tests should be interpreted in light of patient's hydration status.

plus

Urine
specific gravity
<1.030

Urine
specific gravity
<1.035[†]

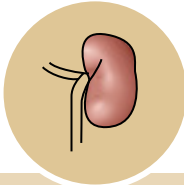
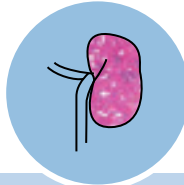
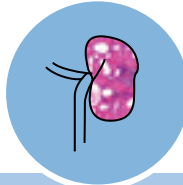
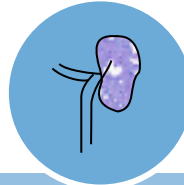


1.030	Canine	1.008
1.035	Feline	1.008

[†]Note that some cats can produce hypersthenuric urine in the face of renal azotemia.

Diagnosing, Staging, and Treating Chronic Kidney Disease in Dogs and Cats

Step 2: Stage CKD

				
	Stage 1 No azotemia (Normal creatinine)	Stage 2 Mild azotemia (Normal or mildly elevated creatinine)	Stage 3 Moderate azotemia	Stage 4 Severe azotemia
Canine				
Creininine in mg/dL Stage based on	Less than 1.4 (125 µmol/L)	1.4–2.8 (125–250 µmol/L)	2.9–5.0 (251–440 µmol/L)	Greater than 5.0 (440 µmol/L)
Feline				
Creininine in mg/dL Stage based on	Less than 1.6 (140 µmol/L)	1.6–2.8 (140–250 µmol/L)	2.9–5.0 (251–440 µmol/L)	Greater than 5.0 (440 µmol/L)
SDMA* in µg/dL Stage based on stable SDMA	Less than 18	18–35	36–54	Greater than 54
UPC ratio Substage based on proteinuria	Nonproteinuric <0.2	Borderline proteinuric 0.2–0.5	Proteinuric >0.5	
	Nonproteinuric <0.2	Borderline proteinuric 0.2–0.4	Proteinuric >0.4	
Systolic blood pressure in mm Hg Substage based on blood pressure	Normotensive <140 Prehypertensive 140–159 Hypertensive 160–179 Severely hypertensive ≥ 180			

Note: In the case of staging discrepancy between creatinine and SDMA, consider patient muscle mass and retesting both in 2–4 weeks. If values are persistently discordant, consider assigning the patient to the higher stage.

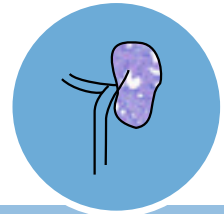
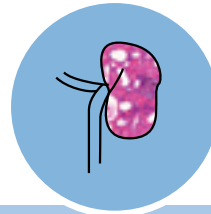
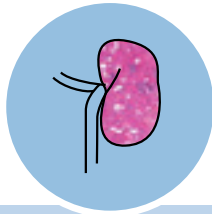
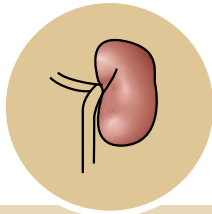
*SDMA = IDEXX SDMA® Test

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Diagnosing, Staging, and Treating Chronic Kidney Disease in Dogs and Cats

Step 3: Treat CKD

Treatment recommendations



Stage 1	Stage 2	Stage 3	Stage 4
Use nephrotoxic drugs with caution	Same as Stage 1	Same as Stage 2	Same as Stage 3
Correct prerenal and postrenal abnormalities	Renal therapeutic diet	Keep phosphorus <5.0 mg/dL (<1.6 mmol/L)	Keep phosphorus <6.0 mg/dL (<1.9 mmol/L)
Fresh water available at all times	Treat hypokalemia in cats	Treat metabolic acidosis	Consider feeding tube for nutritional and hydration support and ease of medicating
Monitor trends in creatinine and SDMA to document stability or progression	Treat inappetence and nausea if present	Consider treatment of anemia	
Investigate for and treat underlying disease and/or complications		Treat vomiting, inappetence, and nausea	
Treat hypertension if systolic blood pressure persistently >160 or evidence of end-organ damage		Increased enteral or subcutaneous fluids may be required to maintain hydration	
Treat persistent proteinuria with renal therapeutic diet and medication (UPC >0.5 in dogs; UPC >0.4 in cats)			
Keep phosphorus <4.6 mg/dL (<1.5 mmol/L)			
If required, use renal therapeutic diet plus phosphate binder			

See www.iris-kidney.com for more detailed staging, therapeutic, and management guidelines.